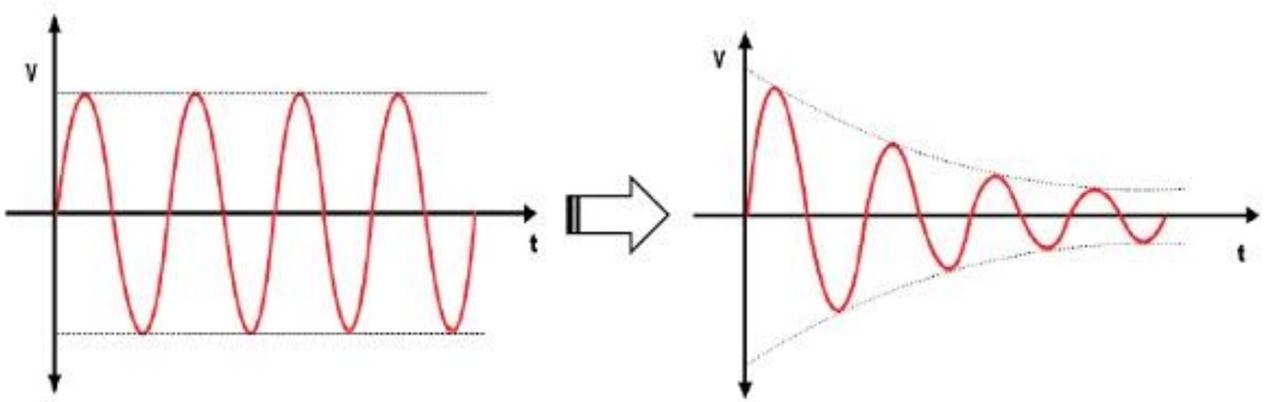
TRANSMISSION IMPAIRMENT

- the signal at the beginning of the medium is not the same as the signal at the end of the medium.
- What is sent is not what is received.
- attenuation, distortion, and noise

Attenuation

- Attenuation means a loss of energy



- Decibel
  - show that a signal has lost or gained strength
  - decibel (dB) measures the relative strengths of two signals or one signal at two different points

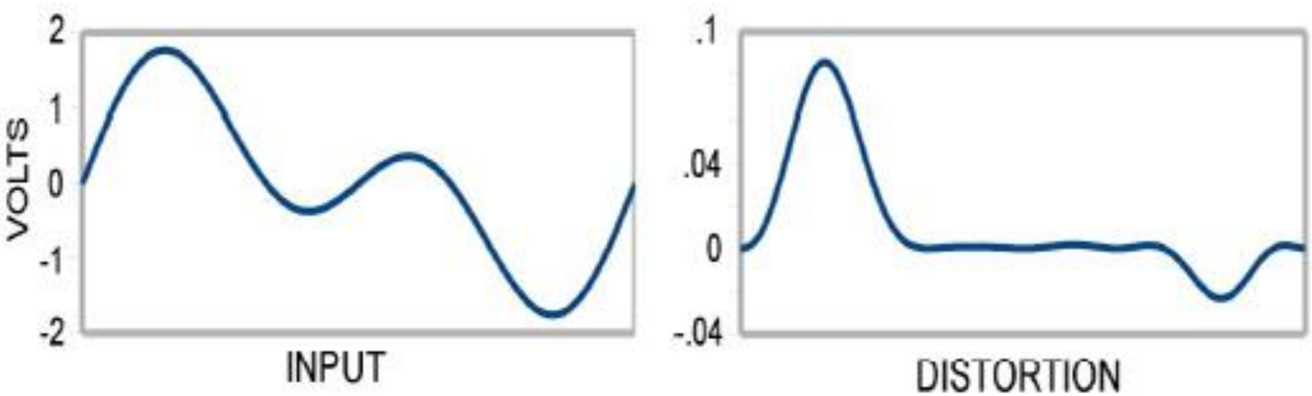
$$\mathbf{dB} = 10 \log_{10} \frac{P_2}{P_1}$$

Suppose a signal travels through a transmission medium and its power is reduced to one-half.

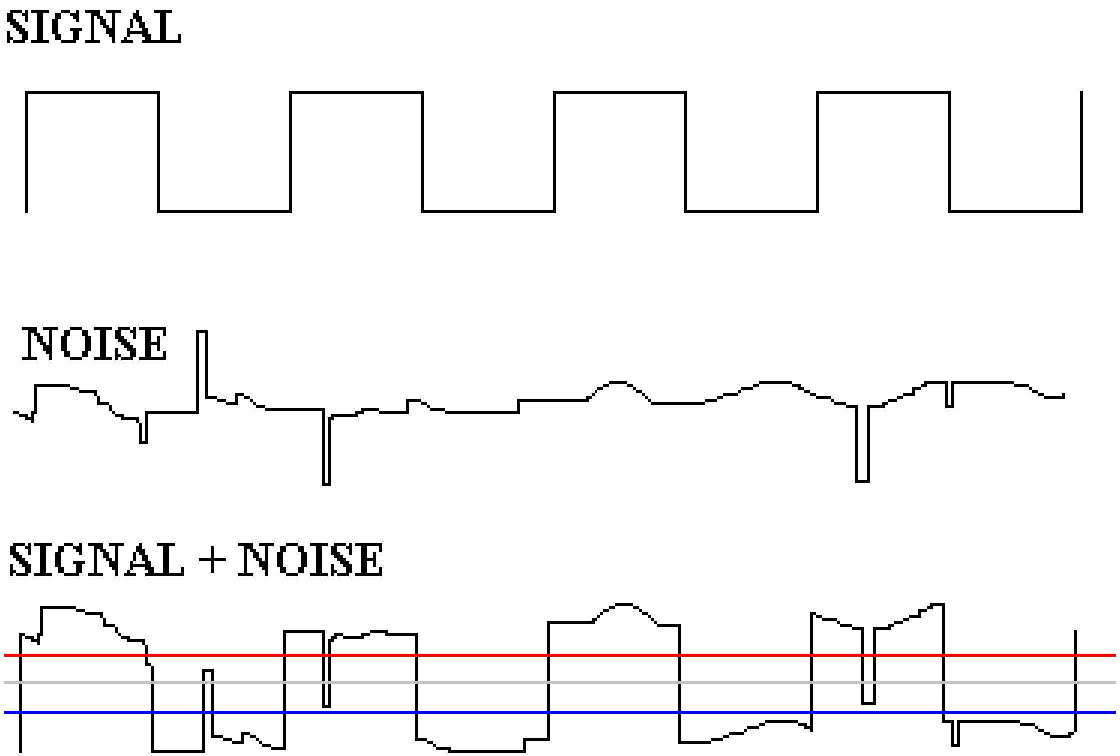
Suppose the signal travels through an amplifier, and its power is increased 10 times.

Distortion

- **Distortion** means that the signal changes its form or shape.



Noise



Signal-to-Noise Ratio (SNR)

- Shannon capacity, to determine the theoretical highest data rate for a noisy channel:

  Capacity = bandwidth X log2 (1 +SNR)

- Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel how is the capacity calculated?

- We measure the performance of a telephone line (4 KHz of bandwidth). When the signal is 10 V, the noise is 5 mV. What is the maximum data rate supported by this telephone line?